

Lesson 21 — Build the Model & Do It Again

Matches LEGO Teacher Portal lesson: "Slide and Share".

Objective (I can...): I can build the loops model and use a repeat block so I don't copy the same block many times. **CS concept:** Loops / repeating actions (repeat N times). A loop is **hiding the mess**: one "repeat 5x" picture stands in for five copied blocks. · **Builds on:** Lesson 20 (an ordered story; now we repeat a chunk of it).

AI Connection (2 min, read aloud): AI gets better by repeating — it sees examples over and over to learn. *Ask the group:* "Why would doing it many times help a computer learn?"

Materials: LEGO CS & AI kit + device per pair. **This is the unit's build lesson** — build the Loops model now and keep it assembled through Lesson 25.

60-minute flow - Warm-up (10): "Do it again" — clap 10 times counting together, then find a shorter way to say it: "repeat 10 times!" Sing a repeating song (a 2-line chorus over and over) and name the repeat. - **Concept + demo (8):** Show 4 copied "move" blocks, then replace them with one move inside a "repeat 4" picture; run both so kids see they match. - **Build/Code (25):** Pairs **build the Loops model** (the unit's one build), then code a repeated action with a repeat picture (spin, pause, repeat 5x). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change the repeat number; guess how much longer it runs before you press Run. - **Share + tidy-up (10):** Pairs show their loop and say the count out loud. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator counts the repeats out loud. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a palette card highlighting the repeat picture. *Extension / fast-finisher* — put two blocks inside the loop. *Reteach* — count the repeats out loud together on the first run.

Exit check: Pair swaps copied blocks for a working repeat loop and can say the count. **At-home (optional):** Find something you do over and over; say it as "repeat ___ times." ~5 min.
